parasites have complicated life cycles, often spread across vertebrate and invertebrate hosts and inhabiting multiple tissues therein. A roadmap of when particular apicoplast capacities are essential in each host for each parasite is gradually being pieced together. For instance, apicoplast fatty acid biosynthesis is essential to *Toxoplasma* parasites living in mice. However, in malaria parasites of mice apicoplast fatty acid biosynthesis is dispensable in the blood phase but essential in the liver phase. In human malaria parasites the isoprenoid precursor pathway is the only essential apicoplast function when the parasites inhabit our red blood cells.

Can we kill it? The apicoplast is essentially a reduced cyanobacterium living inside the parasite. Cyanobacteria are Gram-negative bacteria and sensitive to many antibiotics that target prokaryotic metabolism, and malaria parasites succumb to common antibacterials like ciprofloxacin, clindamycin and doxycycline, the latter being widely used as a malaria prophylactic. Initial suggestions that herbicides inhibit plant-like fatty acid biosynthesis pathways in malaria apicoplasts have proved unfounded. However, herbicidal antibacterials inhibiting apicoplast isoprenoid precursor synthesis are being pursued as novel antimalarials.

Where can I find out more?

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Sniffing patterns uncover implicit memory for undetected odors

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Consciously undetected events are represented at the sensory-motor level and in the neurons of sensory-motor control, for example, consciously undetected visual targets drive eye movements [1] and neural activity [2]. Olfaction offers an opportunity to investigate processing of undetected stimuli through measurements of the sniff-response: odorant-specific modulations of nasal airflow [3–6]. Here, we report evidence that consciously undetected odorants modulate sniffing in a predicted manner. Moreover, in our study we observed that sniff-modulations recurred at least 10 seconds after the onset of an undetected odor, implying that information which was not consciously perceived was nevertheless maintained in memory, available for future decision making.

To test the hypothesis that odors modulate sniff duration in the absence of conscious detection, we measured sniffs in 27 subjects (see the on-line Supplemental Information) during a maximum-likelihood, adaptive staircase olfactory detection task involving a forced-choice between two alternatives. Each trial entailed consecutive presentation of two jars (~10 s between jars), one containing an odorant diluted in mineral oil (*odor*), and the other containing mineral oil only (*blank*), counterbalanced for order. Participants sniffed each jar once and determined which contained an odor. Estimates of conscious perception were based on detection, and estimates of sensory-motor performance on concurrent precise sniff measurement.

Given expected canceling effects between trials where blank preceded odor and odor preceded blank (see the Supplemental Information), we analyzed these trials separately. For trials where blank preceded odor, an analysis of variance (ANOVA) with conditions of *odor* (odor/blank jar) and *accuracy* (correct/incorrect verbal report) revealed a main effect of *odor* ($F(1,26) = 30.37$, $p < 0.00005$), reflecting longer sniffs for blank than odor, no effect of *accuracy* ($F(1,26) = 0.01$, $p > 0.91$), and a significant interaction between *odor* and *accuracy* ($F(1,26) = 4.90$, $p < 0.05$), reflecting a larger difference between odor and blank in correct vs. incorrect trials. Follow-up tests revealed that sniff duration was longer for blank vs. odor for correct (blank = 2491.3 ± 705.2 ms, odor = 2312.4 ± 594.2 ms, $t(26) = 6.2$, $p < 0.0001$) and critically, also for incorrect trials (blank = 2450.0.4 ± 648.9 ms, odor = 2346.3 ± 648.7 ms, $t(26) = 3.4$, $p < 0.005$; Figure S1A), implying olfactory sensory-motor adjustments for events that were not consciously perceived (incorrect trials).

Given that in the above analysis blank was always first, it does not discriminate an effect of odor from an effect of order (Figure S1). To address this, we first analyzed the responses to the first sniff alone, thus avoiding order. An ANOVA on *odor* (odor/blank) and *accuracy* (correct/incorrect) revealed no effect of *odor* ($F(1,26) = 0.11$, $p > 0.74$), no effect of *accuracy* ($F(1,26) = 1.71$, $p > 0.20$) but a significant interaction ($F(1,26) = 11.52$, $p < 0.005$), reflecting longer sniff duration for blank vs. odor in correct trials (odor = 2417.2 ± 589.2 ms, blank = 2491.3 ± 635.7 ms, $t(26) = 2.7$, $p < 0.05$) and shorter in incorrect trials (odor = 2540.6 ± 674.1 ms, blank = 2450.0 ± 648.9 ms, $t(26) = 2.2$, $p < 0.05$; Figure 1A).

We next conducted two follow-up experiments in 54 subjects where trials of two consecutive blanks were embedded without participants' knowledge. In contrast to the expectation following an order effect, we found no difference in sniff duration between the two consecutive blanks (8.3 ± 146.9 ms difference, $t(53) = 0.42$, $p > 0.68$), and importantly, sniff duration difference between two consecutive blanks was significantly smaller than the sniff duration difference between blank and odor, in correct (175.9 ± 146.4 ms difference, $t(79) = 4.8$, $p < 0.00001$) and incorrect (108.6 ± 167.6 ms difference, $t(79) = 2.76$, $p < 0.01$; Figure 1B) trials.

Moreover, in an analysis of trials *around threshold* only (see the

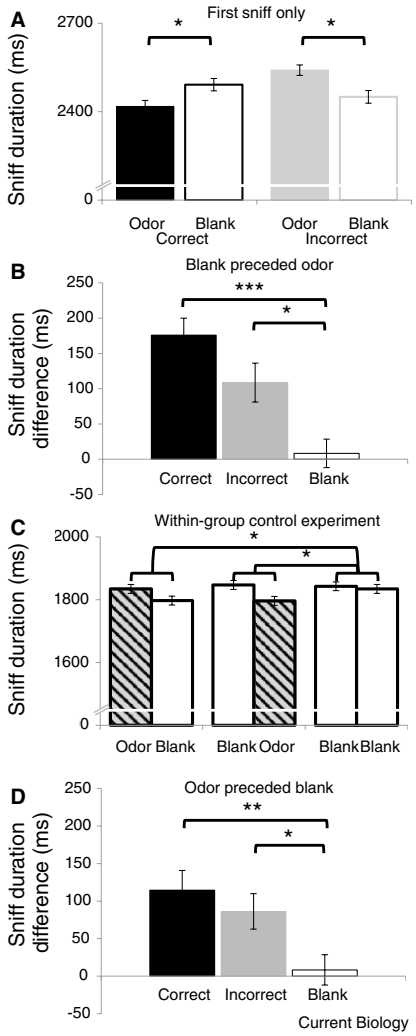


Figure 1. Sniff-modulation following undetected odorants.

(A) Duration for the first sniff only, for correct (odor, solid black; blank, black line) and incorrect (odor, solid gray; blank, gray line) trials (n = 27). (B,D) Duration difference between the first and second sniff in correct (black), incorrect (gray), and blank-blank (outlined) trials, when: (B) blank preceded odor; or (D) odor preceded blank. (C) Sniff duration in the follow-up control experiments for trials where odor preceded blank (left), trials where blank preceded odor (middle) and trials of two consecutive blanks (right), showing no order effect in blank-blank trials. Odor marked in diagonal strips, blank marked in outline (n = 54). Note that the blank-blank comparisons are across-subjects in panels (B,D), and within-subjects in panels (A,C). Error bars are standard error of the mean. ***p < 0.0005, **p < 0.005, *p < 0.05. See also Figure S1 and Tables S1 and S2 in the Supplemental Information.

Supplemental Information), an ANOVA on *order* (1st/2nd jar) and *jar contents* (odor-blank/blank-odor/blank-blank) revealed no effect of *order* ($F(1,53) = 3.12$, $p > 0.09$), no effect of *jar contents* ($F(2,106) = 2.53$, $p > 0.09$), and a significant interaction ($F(2,106) = 2.23$, $p < 0.05$), reflecting that sniff duration difference in blank-blank trials was significantly smaller than sniff duration difference in odor-blank trials ($t(53) = 2.07$, $p < 0.05$) and in blank-odor trials ($t(53) = 2.59$, $p < 0.05$; Figure 1C). Together, these analyses implicate order-independent odor-modulation of sniffing for undetected odors.

Next, we analysed trials where odor preceded blank. An ANOVA on *odor* (odor/blank jar) and *accuracy* (correct/incorrect verbal report) again revealed a main effect of *odor* ($F(1,26) = 18.13$, $p < 0.0005$) reflecting longer sniff for odor vs. blank, a main effect of *accuracy* ($F(1,26) = 13.46$, $p < 0.005$) reflecting a shorter sniff for correct vs. incorrect trials, and no interaction ($F(1,26) = 0.19$, $p > 0.91$). Follow-up tests revealed that sniff duration was significantly longer for odor vs. blank for correct (odor = 2421.3 ± 593.2 ms, blank = 2306.7 ± 591.4 ms, $t(26) = 3.7$, $p < 0.001$) and incorrect (odor = 2542.3 ± 675.0 ms, blank = 2456.0 ± 673.0 ms, $t(26) = 3.1$, $p < 0.005$; Figure S1B) trials.

Moreover, the sniff duration difference between two consecutive blanks (8.3 ± 146.9 ms difference) was significantly smaller than the sniff duration difference between odor and blank, in correct (114.6 ± 159.6 ms difference, $t(79) = 3.0$, $p < 0.005$) and incorrect (86.3 ± 143.3 ms difference, $t(79) = 2.3$, $p < 0.05$; Figure 1D) trials. The significant difference in sniff duration between odor and blank during incorrect trials, which was not observed for two consecutive blanks, implies that although the odor was perceptually undetected, the information about the odor was retained for at least ~10 seconds, and influenced the following sniff of blank. This sniff modulation of the following blank suggests that undetected sensory events influence later sensory motor responses.

In olfaction, undetected odors can induce brain activation and influence behavior. The current results imply that odors can modulate the sniff response despite a genuine lack of conscious detection. This indicates that, like in other modalities [1], the olfactory sensory-motor apparatus is privileged to information unavailable to the conscious behavioral response. This is consistent with the notion that when sensory information is near sensory noise, its representation is limited to 'lower-order' mechanisms such as sensory-motor loops, and only after crossing a threshold, sensory information becomes available to 'higher-order' mechanisms involving conscious awareness [7]. Studies suggested various temporal ranges for the retention of undetected sensory events [8–10]. The current results imply that unperceived olfactory information is retained in memory for at least 10 seconds. This suggests that either olfactory-motor loops have an independent memory buffer, or that consciously undetected olfactory information is represented and retained beyond sensory-motor loops alone.

Supplemental Information

Supplemental Information includes experimental procedures, data, one figure and two tables and can be found with this article online at http://dx.doi.org/10.1016/j.cub.2014.02.004.

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